

Section: **3.1 – STRENGTH OF METALLIC PARTS**
Chapter: **3150**
Title: **PLATE/SHELL BUCKLING**
Revision: **0.1**
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1 INTRODUCTION

1.1 OBJECT

This chapter describes the methodology proposed for calculating the buckling onset of rectangular plane plates or curved shells made of isotropic metallic materials. Two sets of methods are presented:

- ✓ Pre-sizing methods

To be used in initial phases of the design process. They are based in exact solutions or approximations found in literature or engineering handbooks.

- ✓ Methods for final strength check

To be used in final sizing or in certification strength justification. They are based in numerical solutions, have a higher computational cost and shall be utilized only using validated software implementations.

1.2 SCOPE AND LIMITATIONS

Buckling of metallic plates/shells. Limitation of pre-sizing methods:

- ✓ Rectangular shape
- ✓ Plate/shell simply supported or clamped on all its perimeter
- ✓ Other limitations are listed in the dedicated sub-chapter

In any case, the post-buckling behavior of the plate/shell shall not be discussed here.

2 METHODS FOR PRE-SIZING

The pre-sizing methods consist of analytical solutions, handbook methods and available COTS software (such as ESDU programs) with specific simplifications and resulting advantages and disadvantages.

Examples of sources of this kind including solutions for the plate/shell buckling problem under limited boundary conditions and load system include books from Bruhn, Niu, the Roarks and engineering handbooks as the ESDU and HSB sheets.

2.1 RECTANGULAR FLAT PLATES

Elastic buckling under compression, shear and in-plane bending

Recommended references to obtain the elastic buckling stresses of rectangular flat plates of isotropic elastic materials under uniaxial loads are the following:

- ✓ Buckling under uniaxial compression HSB 45111-02 Is. C
- ✓ Buckling under shear loads HSB 45112-01 Is. D
- ✓ Buckling under in-plane bending HSB 45111-04 Is. B

Typical expressions to obtain the critical buckling stresses from material properties (E , ν), plate geometry (a , b , t) and factors (k) with normalized solutions in above mentioned references are:

$$\sigma_{cr} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1 - \nu^2)} \cdot \left(\frac{t}{b}\right)^2 \quad \tau_{cr} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1 - \nu^2)} \cdot \left(\frac{t}{b}\right)^2$$

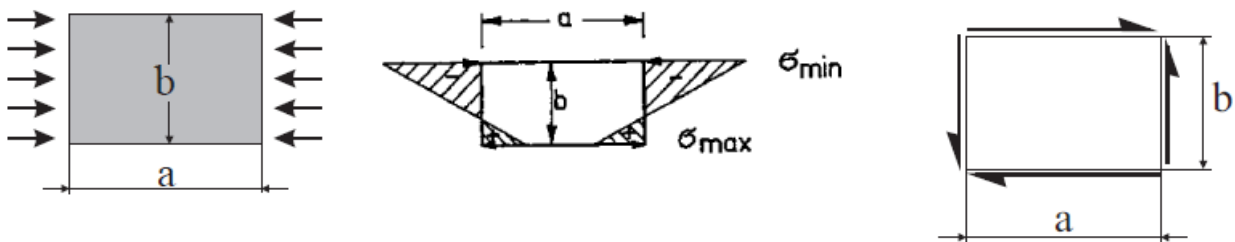


Figure 1. Formulas for calculation of critical buckling stresses (see Ref. 3, Ref. 4, Ref. 5)

2.2 CYLINDRICAL SHELLS

Classical literature and engineering sheets as ESDU and HSB series present closed form solutions and/or graphical approximations for calculating the buckling levels of cylindrical shells subjected to different types of loads and support conditions.

Buckling of narrow curved shells

Recommended references to obtain the elastic buckling stresses of rectangular plates of isotropic elastic materials under uniaxial loads are the following:

- | | |
|--|--------------------|
| ✓ Buckling under longitudinal compression | HSB 45411-01 Is. B |
| ✓ Buckling under shear loads | HSB 45412-01 Is. C |
| ✓ Buckling under in-plane bending | HSB 45111-02 Is. B |
| ✓ Buckling under pressure and longitudinal compression | HSB 45413-02 Is. B |

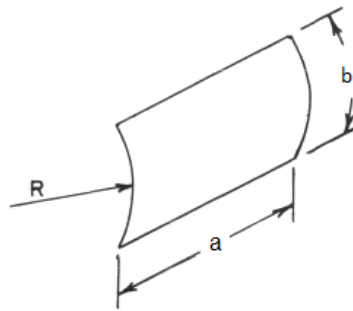


Figure 2. Typical "narrow" curved shell (with $a > b$)

Buckling of wide shells or cylinders

The following methodologies shall be used to calculate the buckling levels of wide cylindrical shells (axial length of the shell between supports small compared with its circumferential length):

- Axial compression ($N_x = \sigma_x \cdot t$): Use Bruhn's Fig. C8.7
- Pressure and circumferential/hoop stresses ($N_y = \sigma_y \cdot t = p \cdot R$)....
 - External radial pressure hoop compression ($p < 0$): Use Bruhn's Fig. C8.16
 - Increase of axial buckling stress due to internal pressure ($p > 0$):

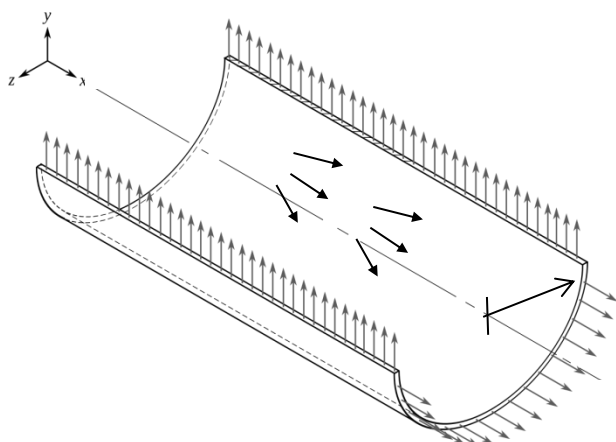


Figure 3. Wide cylindrical shell under axial and pressure/hoop stresses

- Shear:

Use Bruhn's Fig. C8.18

2.3 PLASTIC CORRECTION

From certain thickness to width ratios, previous solutions may give critical buckling stresses near or beyond the yield stress of the material. Those buckling stresses shall have to be corrected due to the inelastic behaviour of the material from vicinities of the yield point and above.

The proposed method to perform this correction is that suggested in Niu's book (Ref. 2), in its chapter §11.2. The linear elastic stress calculated previously shall be multiplied by the factor η_p to be calculated from the stress-strain curve parameters of the material.

$$F_{cr} = \frac{k \eta_p \pi^2 E}{12(1 - \mu^2)} \left(\frac{t}{b}\right)^2$$

Compression stress:

$$\eta_c = \left(\frac{E_t}{E}\right)^{\frac{1}{2}} = \left[\frac{1}{1 + \left(\frac{0.002 E n}{F_{cy}}\right) \left(\frac{F_{c,cr}}{F_{cy}}\right)^{n-1}} \right]^{\frac{1}{2}}$$

Shear stress:

$$\eta_s = \left(\frac{G_t}{G}\right)^{\frac{1}{2}} = \left[\frac{1}{1 + \left(\frac{0.002 G n}{F_{sy}}\right) \left(\frac{F_{s,cr}}{F_{sy}}\right)^{n-1}} \right]^{\frac{1}{2}}$$

Figure 4. Correction factor to buckling stresses due in inelastic materials, per Niu's book §11.2

Notice the correction factor η_p must be calculated at the actual/real buckling stress, thus the problem results in a non-linear equations system that has to be solved iteratively (or using graphical solution in Niu's book).

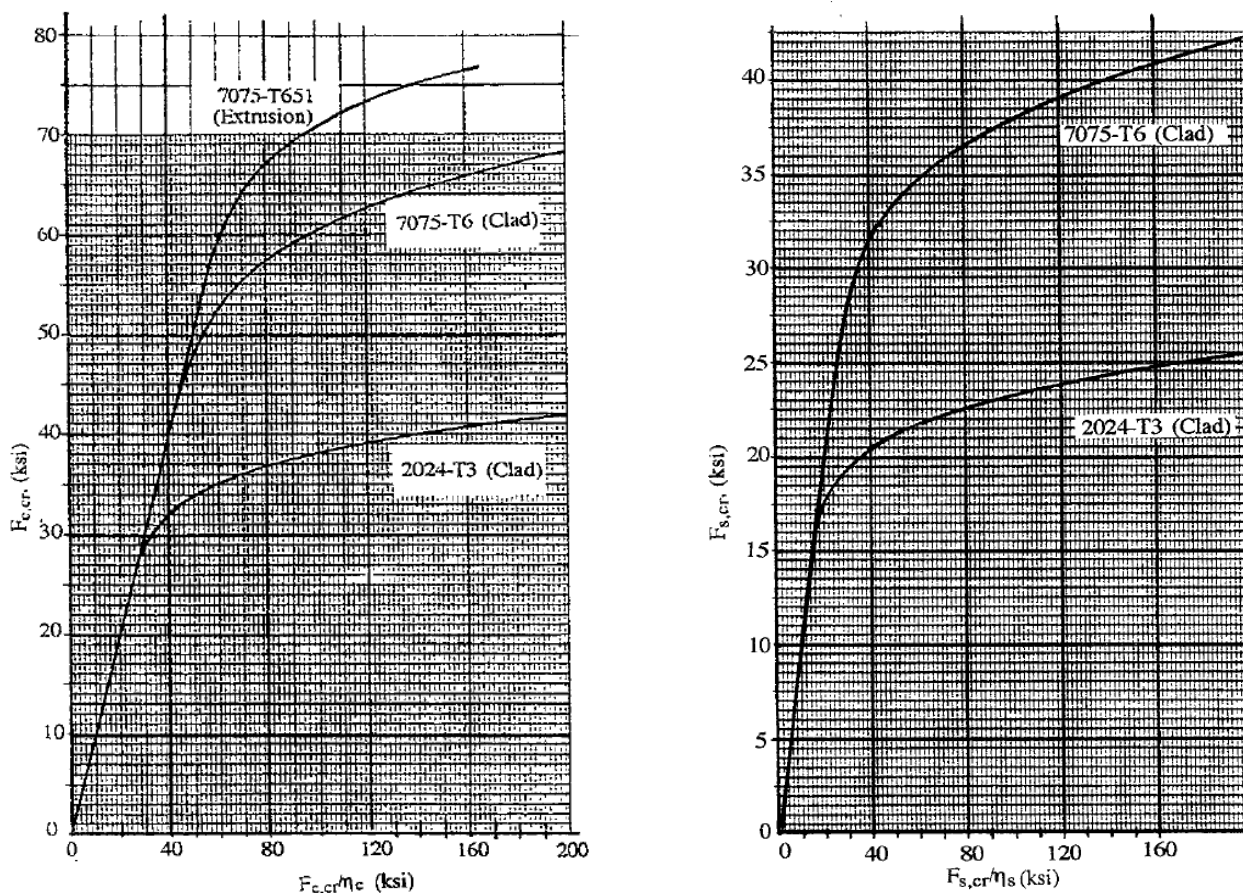


Figure 5. Corrected buckling compression and shear stresses obtained from linear-elastic solutions, Figures 11.2.4 and 11.2.5 in Niu's book

2.4 BUCKLING UNDER COMBINED LOADS

Abovementioned references present typical interaction formulas used to obtain the safety margin vs buckling stability in multi-axial stress states. The proposed failure law is:

$$R_c + R_f^{1.75} + R_s^2 = 1$$

...where R_c , R_f and R_s are the failure indices in uni-axial loads for axial longitudinal compression, in-plane bending (flexure) and shear respectively, i.e.:

$$\begin{aligned} R_c &= 1 / BL && \text{...if obtained from the explicit solution available for bi-axial loads} \\ &= R_x + R_y && \text{...from uni-axial components, with } R_x = -N_x / N_{xb}, R_y = -N_y / N_{yb} \end{aligned}$$

(*) When any of the axial forces N_x or N_y is positive tension the recommended (conservative) approach is to make the corresponding failure index equal to 0 in the interaction formula.

$$R_f = M / M_b$$

$$R_s = N_{xy} / N_{xyb}$$

That expression can be expressed introducing the Reserve Factor for performing the strength check of a general load case:

$$RF \cdot R_c + (RF \cdot R_f)^{1.75} + (RF \cdot R_s)^2 = 1$$

If the in-plane applied bending M is not zero, the Reserve Factor shall have to be solved numerically.

3 METHODS FOR FINAL STRENGTH CHECK

The methods presented in previous section are generally, conservative, and indeed may still be used in final sizing steps, or in certification justification, provided their limitations have not been exceeded. In addition to those inherent limitations of the analytical methods, under some circumstances, more refined methodologies shall be needed to evaluate the accurately the buckling levels of the part. These circumstances may include short static margins, weight saving campaigns, loads exceedances in final load loops, ..., among others.

At Airbus DS engineering office, the available implementations of these more refined tools (all supported in numerical methods) include:

Energy (Rayleigh-Ritz) methods

These methods compute the total energy of the structural system as a sum of the deformation energy of the structure plus the potential energy of the applied loads. The deformed shape is approximated by series of functions compatible with the boundary conditions, with the coefficients of these functions unknown. The solution for the coefficients is obtained deriving the total energy and making it equal to zero for searching its minimum. The eigenvalues and eigenvectors of the resulting linear system of equations are the buckling levels and buckling modes.

Current implementations cover rectangular plane plates, un-stiffened or stiffened with horizontal or vertical stringers. They are capable to detect local and global buckling modes, and to model rotational constraints at supports (beyond the simple-supports conditions).

See Ref. 10 for details.

Implementations for curved panels are currently NOT available at Airbus DS.

Finite Differences methods

To Do: Evaluate and test the available implementation developed by the DLR and integrated in Strength 2000, capable to deal with curved stiffened panels

Detailed Finite Elements Models

For specially complicated parts the use of FEM analysis is the last and more sophisticated tool to solve the buckling analysis problem.

Rules and recommendations for adequate preparation and processing FEM analyses are out of the scope of this chapter. See Volume 2 of the Stress Methods Manual for further information.

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Length of the plate/shell (dimension in X-longitudinal direction)
b	mm / -	Width of the plate/shell (dimension in Y-transversal direction) or, as subscript, relative to buckling
BL	-	Buckling level, ratio between buckling load and applied load
c	-	As subscript, relative to compression
cr	-	As subscript, relative to critical buckling stress
COTS	-	Commercial off-the-shelf
E	MPa	Young modulus of the material
f	-	As subscript, relative to in-plane bending
I	mm ⁴	Bending moment of inertia
M _b	Nmm	In-plane bending moment of the plate at buckling onset
N _{xb}	N/mm	Compression buckling load-flow (force per unit width) in X direction
N _{xyb}	N/mm	Shear buckling load-flow (force per unit width)
N _{yb}	N/mm	Compression buckling load-flow (force per unit width) in Y direction
q	N/mm	In plane shear force per unit width
R	mm / -	Curvature radius of the shell (around an axis parallel to X direction) or, form context, failure index, i.e. ratio between uni-axial load and allowable load
RF	-	Reserve Factor, ratio between allowable load and applied load
s / xy	-	As subscripts, relative to in-plane shear
t	mm	Thickness of the plate/shell
η _p	-	Plastic correction factor to linear-elastic buckling stress
ν	-	Poisson's ratio of the material (typ. 0.33 for aluminium alloys, 0.31 for Titanium alloys and 0.28 for steel alloys)
σ	MPa	Axial stress
τ	MPa	In-plane mean shear stress

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	[Michael C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 3	HSB 45111-02	Is. C	Buckling of flat, rectangular plates under uniaxial compression
Ref. 4	HSB 45112-01	Is. D	Shear buckling of flat, rectangular plates
Ref. 5	HSB 45111-04	Is. B	Buckling of flat, rectangular plates under bending and compression
Ref. 6	HSB 45111-01	Is. B	Buckling of curved plate strips under compression loading
Ref. 7	HSB 45412-01	Is. C	Buckling of curved plate strips under shear loading
Ref. 8	HSB 45411-02	Is. B	Buckling of curved plate strips under bending loading
Ref. 9	HSB 45413-02	Is. B	Buckling of curved plate strips under combined axial compression and internal pressure loading as well as shear and internal pressure loading
Ref. 10	NT-T-ADP-06002	A	Manual teórico del programa PANPA, versión 5.0

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
NASTRAN (MSC-Software)	2005	Multidisciplinary structural analysis software based in the Finite Elements method
PANPA (Airbus DS)	5.4	Buckling of rectangular panels (energy Rayleigh-Ritz method)

Table 1. *Software programs referenced in this chapter*

DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Fernass Daoud		X
Fatigue & Damage Tolerance GET (TEASA-TL1)	Javier Gomez-Escalonilla Martin		
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
Stress Combat MAN (TEASA-TL3)	Sebastian Rönz	X	
Stress Combat, Aerostructures, A400M and Technologies GET (TEASA-TL4)	Sofia Ponce Borrero	X	
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Structural Repair GET (TEASA-TL6)	Jose Luis Melchor Martin de la Sierra	X	
Fatigue & Damage Tolerance MAN (TEASA-TL7)	Patrick Schneider		

Table 2. *Distribution list (of electronic document, ahead of public release)*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 04/05/2019	G. Baños	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEASA)	Miguel Ángel Calero XX/XX/2018	Bernhard Mayer XX/XX/2018
Stress Methods (TEASP-TL3)	G. Baños 04/05/2019	Florian Glock XX/XX/2018

Table 4. *Approval signatures table for last revision of this chapter*